

Structural Diagnostics for Latent World Models: A LunarLander Case Study of Offline Checkpoint Selection for MPC and Model-based Actor-Critic Policy

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Abstract

We study how to predict the downstream closed-loop performance of a learned latent world model directly from validation-time diagnostics, without running expensive real-environment policy evaluations. Choosing the right checkpoint from a training run of a learned latent world model is difficult: validation loss and multi-step prediction RMSE keep improving long after closed-loop performance has collapsed. To address this gap, we present a suite of structural validation-time diagnostics for latent world models, drawn from optimal-control theory, and apply them to Gymnasium’s LunarLander-v3 environment, a small but non-trivial testbed well suited to direct analysis of the learned latent. We train a Dreamer-style Recurrent State Space Model (RSSM), treat per-checkpoint CEM-MPC return as the oracle for closed-loop quality, and evaluate 40 metrics against this oracle, including Jacobian-based controllability and observability and four reward/observation subspace-alignment scores. The strongest single predictive metric is the Reward Observability Fraction (ROF), which measures the reward predictor’s dependence on the observable subspace. We combine ROF with three structural metrics into a single-number offline checkpoint-selection score, the Composite Reward–Observability Fraction (CROF). The CROF-selected world model trains a model-based A2C policy that beats a fairly evaluated model-free A2C baseline by ~ 24 return points while using $\sim 65\times$ fewer real-environment interactions, and the same world model also drives a strong zero-shot CEM-MPC policy. All experimental claims are on LunarLander-v3 and cross-environment validation is left to future work. Code and data: https://github.com/nsmoly/LunarLander_RSSM.

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1 Introduction

Model-based reinforcement learning promises to reduce the number of real-environment interactions needed to learn effective policies by building a predictive model of the environment—a *world model*—and using it for planning or policy optimization in imagination. The Dreamer or Recurrent State Space Model (RSSM) architecture [5, 4] has become a standard backbone for world models, enabling both online planning via Model Predictive Control (MPC) [5] and policy learning with Reinforcement Learning (RL) through imagined rollouts [4, 6].

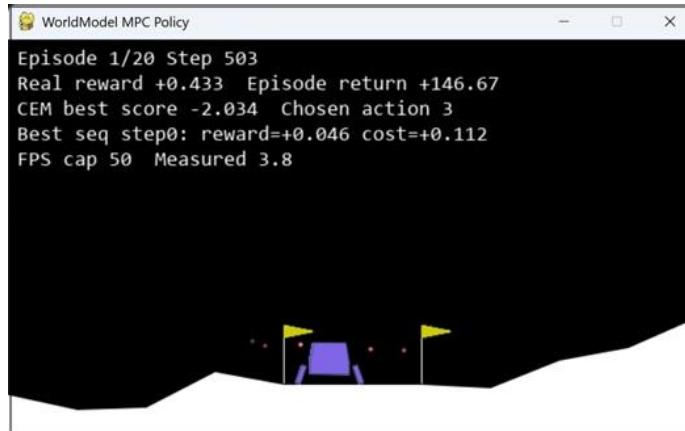


Figure 1: The LunarLander-v3 environment. The agent can use four discrete actions (all engines OFF, left engine ON, main engine ON, right engine ON) and must softly land the lander between the two flags from a randomized initial position and velocity. We use it as a testbed for studying world model quality: its dynamics are non-trivial (rotational, translational) yet its 8-dimensional observation space is small enough to directly analyze the structure of the learned latent space.

A fundamental challenge in model-based RL is *objective mismatch* [8]: the training objective for the world model (prediction accuracy) does not necessarily align with the downstream objective (closed-loop performance). A model with excellent one-step or even multi-step prediction accuracy may still be unsuitable for planning or policy optimization if its learned latent dynamics lack the structural properties needed for these purposes.

In this work, we focus on the LunarLander-v3 Gymnasium environment. This environment has not been the subject of detailed RSSM/Dreamer-style analysis in prior work, yet it has the right combination of properties for a controlled study: non-trivial dynamics with rotational and translational coupling, but a low-dimensional observation and action space that admits direct structural analysis of the learned latent. We address the objective-mismatch question from two complementary angles. First, we develop a suite of diagnostic metrics drawing on classical control theory (controllability, observability, Kalman [7]) that measure the structural adequacy of a learned world model for control. Second, we show experimentally on LunarLander that selecting a world-model checkpoint using these structural metrics leads to substantially better downstream policy performance than standard model-selection criteria, both for MPC and for model-based actor-critic policy trained in imagination.

Our contributions in this case study are:

1. A suite of 40 structural metrics for world-model evaluation, including Jacobian-based controllability/observability analysis (fixed and time-varying linearizations) and four reward/observation subspace-alignment scores (RCF, OCF, ROF, CO overlap). The strongest single metric,

`jac_rof_combined`, reaches Spearman $\rho_s=-0.71$ and $R_q^2=0.52$ against smoothed MPC return.

2. The Composite Reward–Observability Fraction (CROF), a single-number offline checkpoint-selection score that combines ROF with three structural regularizers (controllability rank, observability rank, open-loop observation error). We evaluate two weight variants, CROF-A and CROF-B.
3. Experimental evidence on LunarLander-v3 shows that CROF tracks both zero-shot CEM-MPC returns and downstream A2C training quality, while standard selection criteria (validation loss, multi-step RMSE, sensitivities) instead pick deeply overfit checkpoints. CROF identifies world-model checkpoints within ~ 5 – 15 epochs of the smoothed MPC oracle peak where zero-shot world-model-based CEM-MPC reaches mean return $+166.6$ without any policy training. CROF metric-driven picks yield the strongest and safest world model checkpoints for training A2C policy, which beats the model-free A2C baseline on peak return ($+217.5$ vs. $+193.0$).
4. A data-efficiency comparison showing that the CROF-selected world model trains a model-based A2C policy that beats a fairly evaluated model-free A2C baseline at $\sim 65\times$ less real-environment interaction (181K offline steps vs. 11.8M).

2 Related Work

Our work draws on three areas: world models for control, the objective mismatch problem in model-based RL, and classical control-theoretic analysis of dynamical systems.

2.1 World Models and Model-Based RL

The RSSM architecture was introduced by Hafner et al. in PlaNet [5], which demonstrated that planning in a learned latent space can solve continuous control tasks from pixels. Dreamer [4] extended this by learning an actor-critic within the latent imagination of the world model, and DreamerV3 [6] scaled the approach to over 150 diverse tasks with a single configuration. In parallel, PETS [2] showed that probabilistic ensemble dynamics models combined with MPC can achieve strong sample efficiency. All of these methods train world models by maximizing prediction likelihood (or minimizing reconstruction error) and evaluate them by downstream task performance, without analyzing the structural properties of the learned dynamics that determine planning quality.

2.2 Objective Mismatch

Lambert et al. [8] identified the *objective mismatch* problem: optimizing a dynamics model for one-step prediction accuracy does not guarantee good downstream closed-loop performance. They demonstrated empirically that prediction likelihood is not always correlated with task reward, and proposed re-weighting training objectives as an initial mitigation. Our work proposes an explanation of why this mismatch occurs: prediction accuracy (even multi-step) captures statistical fit, while planning quality depends on the structural alignment between reward-relevant, controllable, and observable subspaces — a geometric property that training losses do not measure.

2.3 Task-Relevant Representations

Several works aim to learn representations that preserve task-relevant information. DeepMDP [3] learns latent spaces that preserve reward and transition structure via bisimulation-inspired losses, providing theoretical guarantees on representation quality. SOLAR [13] learns representations where linear dynamics are accurate, enabling LQR-based planning from images. Agarwal et al. [1] introduced policy similarity metrics and contrastive embeddings that capture behavioral similarity between states. These approaches address the mismatch problem by modifying the training objective. Our work is complementary: rather than changing how models are trained, we provide post-hoc diagnostic validation-time metrics that quantify whether the learned structure supports closed-loop planning, regardless of the training procedure.

2.4 Control-Theoretic Analysis of Learned Systems

The controllability and observability framework originates with Kalman [7], who showed that the rank of the controllability and observability matrices determines whether a linear system can be fully steered and fully observed. Moore [9] introduced balanced realization theory, using SVD of controllability and observability Gramians to identify state dimensions that are simultaneously controllable and observable — the dimensions most relevant for control. Our Controllability-observability overlap (CO) metric is directly inspired by this idea.

Sussillo and Barak [12] applied linearization and fixed-point analysis to trained RNNs in neuroscience, showing that local Jacobian analysis reveals interpretable low-dimensional dynamics in high-dimensional recurrent networks. We adapt this approach to the RSSM setting, using Jacobians not for interpretation but for quantifying structural adequacy for planning.

Novelty of subspace-alignment metrics. Jacobian linearization, controllability/observability matrices, SVD-based rank analysis, and reward gradient computation are all standard tools; the contribution here is their combination as scalar reward/observation alignment fractions (RCF, OCF, ROF) and their use as *post-hoc validation-time diagnostics* for learned world models, predicting closed-loop performance without environment access.

3 Method

3.1 World Model Architecture

We use an RSSM-inspired world model [4, 6] adapted for the LunarLander-v3 environment ($p=8$ observation dimensions, $K=4$ discrete actions). The observation space consists of 6 continuous physics dimensions (horizontal position x , vertical position y , horizontal velocity v_x , vertical velocity v_y , angle θ , angular velocity $\dot{\theta}$) and 2 binary contact indicators (left leg, right leg).

The RSSM core maintains a deterministic hidden state $\mathbf{h}_t \in \mathbb{R}^{d_h}$ ($d_h=256$, single-layer GRU) and a stochastic latent $\mathbf{z}_t \in \mathbb{R}^{d_z}$ ($d_z=16$), giving a full latent state $\mathbf{s}_t = [\mathbf{h}_t; \mathbf{z}_t] \in \mathbb{R}^{272}$. Actions $a_t \in \{0, 1, 2, 3\}$ are one-hot encoded as $\mathbf{u}_t \in \mathbb{R}^4$.

The RSSM transition (prior) is

$$\mathbf{s}_{t+1} = f_{\theta}(\mathbf{s}_t, \mathbf{u}_t) = [\text{GRU}(\mathbf{h}_t, [\mathbf{z}_t; \mathbf{u}_t]); \boldsymbol{\mu}_{\text{prior}}(\mathbf{h}_{t+1})], \quad (1)$$

the posterior encoder is $q_{\theta}(\mathbf{z}_t | \mathbf{h}_t, \mathbf{o}_t)$. Both prior and posterior networks are 2-layer MLPs with LayerNorm and SiLU activations, predicting mean and log-standard-deviation of a diagonal Gaussian. The observation encoder is similarly a 2-layer MLP that maps $\mathbf{o}_t \in \mathbb{R}^8$ to a feature vector before being concatenated with $\mathbf{h}_t$ for the posterior.

Structured decoder with heterogeneous heads. Rather than treating all 8 observation dimensions identically, the decoder uses a shared 2-layer MLP backbone (LayerNorm + SiLU, input $d_h+d_z=272$) feeding four heads: an MSE-regression *physics* head ($\mathbb{R}^6$, continuous states), a BCE *contact* head ($\mathbb{R}^2$ logits, sigmoid at inference), a BCE *done* head ($\mathbb{R}^1$), and a separate 3-layer *reward* MLP ($\mathbb{R}^{272} \rightarrow 256 \rightarrow 256 \rightarrow 1$) that operates directly on $[\mathbf{h}; \mathbf{z}]$ rather than the shared backbone, so the reward predictor develops its own representation without competing with reconstruction for backbone capacity. The observation decoder is $\hat{\mathbf{o}}_t = g_\theta(\mathbf{s}_t)$ (physics concatenated with sigmoid of contact logits); the reward predictor is $\hat{r}_t = \rho_\theta(\mathbf{s}_t)$.

Training loss. The total loss is a weighted sum:

$$\mathcal{L} = w_r \mathcal{L}_{\text{recon}} + w_\rho \mathcal{L}_{\text{rew}} + w_d \mathcal{L}_{\text{done}} + w_{\text{KL}} \mathcal{L}_{\text{KL}}, \quad (2)$$

with weights $w_r=1.0$, $w_\rho=1.2$, $w_d=0.5$, $w_{\text{KL}}=1.0$ (KL free-bits floor $\beta_{\text{KL}}=0.5$). $\mathcal{L}_{\text{recon}}$ combines MSE for physics and BCE for contacts; $\mathcal{L}_{\text{rew}}$ is squared reward error; $\mathcal{L}_{\text{done}}$ is done-flag BCE; $\mathcal{L}_{\text{KL}} = \text{KL}(q_\theta \| p_\theta)$ is the posterior–prior KL divergence.

3.2 MPC via Cross-Entropy Method (CEM)

Since LunarLander-v3 has a discrete action space (4 actions: NOP, left thruster, main engine, right thruster), standard gradient-based planning does not apply. We use the Cross-Entropy Method (CEM) [10] adapted for discrete actions: at each real timestep, CEM iteratively refines a per-step categorical distribution over a planning horizon of $H=25$ steps. Each iteration samples $N=384$ candidate action sequences from the current distribution, rolls each one through the world-model prior open-loop while accumulating discounted reward $\sum_{k=1}^H \gamma^{k-1} \hat{r}_{t+k}$ ($\gamma=0.97$), keeps the top $E=48$ elites, and updates the per-step logits toward the elite action frequencies with smoothing factor $\alpha=0.7$. After $I=4$ such iterations, the first action of the best sequence is executed and the procedure repeats at the next real timestep. The planning horizon $H=25$ matches the rollout horizon used for metric evaluation, so the offline metrics measure exactly the regime CEM operates in.

3.3 Actor-Critic Policy via Latent Imagination

Following Dreamer [4], we train an actor–critic policy (A2C) entirely within the latent imagination of the world model. The actor and critic are compact 3-layer MLPs (hidden dimension is 256) that read the latent state $[\mathbf{h}; \mathbf{z}]$; the actor outputs a categorical distribution over the 4 actions, the critic predicts scalar value estimates. Each training step samples a real observation–action sequence from the replay dataset, warms the RSSM posterior up for 5 steps on the real data, then imagines 15 further steps by rolling out the prior with actions sampled from the actor (no observation correction). Value targets are λ -returns [11] ($\gamma=0.99$, $\lambda=0.95$) computed from the world model’s reward head; the actor is updated by REINFORCE with advantage clipping, the critic by MSE, and entropy is decayed from 0.5 to 0.15 over training. We train for 1000 epochs at learning rate 3×10^{-5} and batch size 256. Crucially, this stage uses *zero additional real-environment interactions*: all learning happens in the learned world model on the same offline dataset used to train it.

3.4 Metrics for World Model Quality

We define metrics in four categories. The full suite is summarized in Table 1.

3.4.1 Validation Loss Decomposition

The weighted total validation loss $\mathcal{L}_{\text{val}}$ and its components ($\mathcal{L}_{\text{KL}}$, $\mathcal{L}_{\text{recon}}$) provide standard training diagnostics. One-step posterior RMSE metrics (`post_obs_rmse`, `post_rew_rmse`) measure closed-loop prediction quality where the encoder corrects the latent state at each step.

3.4.2 Multi-Step Open-Loop Rollout

These metrics directly measure what CEM experiences during planning: multi-step prediction quality without encoder corrections. After 5-step posterior warm-up, the RSSM prior is rolled out for 25 steps using ground-truth actions. All errors are accumulated as squared errors over valid time steps (masking padded/post-terminal positions in the validation sequences) and reduced by a square root, giving true RMSE values. We report:

- **Observation RMSE:** at start (step 1), average, max, and end (step 25) of the rollout (`ol_obs_start/avg/max/end`).
- **Reward RMSE:** at start, average, max, and end (`ol_rew_start/avg/max/end`).
- **Cumulative reward RMSE** (`ol_cumrew_err`): RMSE of the predicted cumulative reward $\sum_{k=1}^H \hat{r}_{t+k}$ against the ground truth, masked over valid sequence positions.

3.4.3 Jacobian-Based Linear Analysis

We linearize the transition, decoder, and reward functions around sampled validation states and apply classical control-theoretic analysis [7].

At a state $\mathbf{s} = [\mathbf{h}; \mathbf{z}]$ with action $\mathbf{u}$, the Jacobians are:

$$A = \left. \frac{\partial f_{\theta}}{\partial \mathbf{s}} \right|_{\mathbf{s}, \mathbf{u}} \in \mathbb{R}^{n \times n}, \quad B = \left. \frac{\partial f_{\theta}}{\partial \mathbf{u}} \right|_{\mathbf{s}, \mathbf{u}} \in \mathbb{R}^{n \times K}, \quad (3)$$

$$C = \left. \frac{\partial g_{\theta}}{\partial \mathbf{s}} \right|_{\mathbf{s}} \in \mathbb{R}^{p \times n}, \quad R = \left. \frac{\partial \rho_{\theta}}{\partial \mathbf{s}} \right|_{\mathbf{s}} \in \mathbb{R}^{1 \times n}. \quad (4)$$

Fixed vs. time-varying linearization. In *fixed* linearization (`jac_` prefix), Jacobians A, B, C, R are computed once at a sampled state, and the controllability/observability matrices use powers A^k : $\mathcal{C}_H = [B, AB, A^2B, \dots, A^{H-1}B]$, $\mathcal{O}_H = [C; CA; CA^2; \dots; CA^{H-1}]$. In *time-varying* linearization (`jac_dyn_` prefix), fresh Jacobians A_k, B_k, C_k, R_k are computed at each step of an actual latent rollout through the prior:

$$\mathcal{C}_H = [B_0, A_1B_0, A_2A_1B_0, \dots, B_1, A_2B_1, \dots, B_{H-1}], \quad (5)$$

$$\mathcal{O}_H = [C_0; C_1A_0; C_2A_1A_0; \dots; C_{H-1} \prod_j A_j]. \quad (6)$$

A priori, time-varying linearization should better capture the nonlinear dynamics actually experienced during CEM planning. Empirically, however, we found that the *fixed* formulation is the stronger predictor of MPC performance for this task (Section 5.3). We retain both as supported variants but adopt the fixed Jacobian as the primary reported metric.

Subspace alignment metrics. From the SVD of the controllability matrix $\mathcal{C}_H = U_c \Sigma_c V_c^\top$ and observability matrix $\mathcal{O}_H = U_o \Sigma_o V_o^\top$, we define effective ranks using a threshold of 10^{-3} relative to the largest singular value. Let $U_c^{(k)} \in \mathbb{R}^{n \times k_c}$ be the controllable subspace basis, $V_o^{(k)} \in \mathbb{R}^{n \times k_o}$ the observable subspace basis, and $\mathbf{r} = R^\top \in \mathbb{R}^n$ the reward gradient. We use the Frobenius norm $\|\cdot\|_F$ for matrices and the Euclidean (ℓ_2) norm $\|\cdot\|$ for vectors throughout. The four subspace-alignment scores are:

- **Reward Controllability Fraction (RCF):** fraction of the reward gradient’s energy in the controllable subspace.

$$\text{RCF} = \frac{\|U_c^{(k)\top} \mathbf{r}\|^2}{\|\mathbf{r}\|^2} \in [0, 1]. \quad (7)$$

- **Observation Controllability Fraction (OCF):** fraction of the decoder’s sensitivity in the controllable subspace.

$$\text{OCF} = \frac{\|C U_c^{(k)}\|_F^2}{\|C\|_F^2} \in [0, 1]. \quad (8)$$

- **Reward Observability Fraction (ROF):** fraction of the reward gradient’s energy in the observable subspace.

$$\text{ROF} = \frac{\|V_o^{(k)\top} \mathbf{r}\|^2}{\|\mathbf{r}\|^2} \in [0, 1]. \quad (9)$$

Lower ROF = better world model for open-loop planning (rationale below).

- **Controllable–Observable Overlap (CO overlap):** subspace alignment between controllable and observable directions.

$$\text{CO_overlap} = \frac{\|U_c^{(k)\top} V_o^{(k)}\|_F^2}{\min(k_c, k_o)} \in [0, 1]. \quad (10)$$

Connection to control theory. Controllability and observability are classical prerequisites for optimal control [7]: action sequences must be able to steer the latent state, and the encoder/decoder pathway must be able to track those changes. Both properties carry over directly to MPC and latent-imagination RL, which roll the world model open-loop through its prior. ROF is particularly informative in this regime because it quantifies how much the reward predictor depends on the observable subspace—the same subspace that drifts during open-loop imagination, since no real observations arrive to anchor it. High ROF means the reward gradient is concentrated in directions the encoder/decoder must constantly correct, so reward predictions become unreliable as soon as the rollout leaves the posterior regime. Low ROF means the reward predictor reads from latent directions that need no observation grounding, which is the regime in which CEM-MPC and latent-imagination A2C operate.

Curated state sampling. Each Jacobian-based metric is averaged over N_J sampled validation states. LunarLander rollouts cluster naturally into successful landings (return $\geq +100$) and failure trajectories (crash or fly-out, return ≤ -100). Uniform random sampling therefore produces mixed batches whose composition varies non-trivially with the seed and dilutes the signal. We instead compute each Jacobian metric twice: once on a batch of “good” (high-return) sequences and once

on a batch of “bad” (low-return) sequences, written `jac_rof` and `jac_rof_bad` for the ROF case. We additionally report their convex combination

$$\text{jac_rof_combined} = \alpha \cdot \text{jac_rof} + (1 - \alpha) \cdot \text{jac_rof_bad}, \quad (11)$$

with $\alpha=0.5$ fixed throughout the paper. This makes the metric *state-aware*: a model is rewarded both for clean reward representation along successful trajectories and for remaining structurally consistent on failure trajectories that MPC must discriminate against. Empirically `jac_rof_combined` is the strongest single predictor we measured (Section 5.2), and a sweep over α confirms a broad optimum near $\alpha \in [0.3, 0.5]$.

3.4.4 Empirical Sensitivity Metrics

These use finite-difference estimation rather than Jacobians:

- Action sensitivity (`emp_C`): mean pairwise distance between next states produced by different actions.
- State perturbation sensitivity (`emp_0`): response of transitions to perturbations in $\mathbf{z}$.
- Lipschitz constant (`emp_L`): *maximum* over sampled state pairs of the ratio $\|f_\theta(\mathbf{s}_1, \mathbf{u}) - f_\theta(\mathbf{s}_2, \mathbf{u})\| / \|\mathbf{s}_1 - \mathbf{s}_2\|$ (max captures localized sensitivity blow-ups that the mean would dilute).

All sensitivity computations are seeded by a local `torch.Generator` for bit-for-bit reproducibility across checkpoints.

3.5 Composite Reward–Observability Fraction (CROF)

`jac_rof_combined` is the single strongest individual predictor of MPC performance based on correlation analysis. However, its raw minimum across training does not necessarily coincide with the smoothed MPC peak: in particular, ROF is not bounded away from zero at early, undertrained checkpoints (where the latent space is barely organized at all), creating a false-positive risk for checkpoint selection. To mitigate this, we introduce the **Composite Reward–Observability Fraction (CROF)**, which combines ROF with complementary structural and prediction-quality terms.

We define two variants that share the same four components but use different relative weights:

$$\text{CROF-A} = \widetilde{\text{ROF}} + 1.0 \cdot (1 - \widetilde{k}_c) + 1.0 \cdot (1 - \widetilde{k}_o) + 1.0 \cdot \widetilde{e}_{\text{obs}}, \quad (12)$$

$$\text{CROF-B} = \widetilde{\text{ROF}} + 0.5 \cdot (1 - \widetilde{k}_c) + 0.5 \cdot (1 - \widetilde{k}_o) + 0.5 \cdot \widetilde{e}_{\text{obs}}, \quad (13)$$

where $\widetilde{(\cdot)}$ denotes min–max normalization across all checkpoints, $\text{ROF} = \text{jac_rof_combined}$ from (11), $k_c = \text{jac_ctrl_rank}$, $k_o = \text{jac_obs_rank}$, and $e_{\text{obs}} = \text{ol_obs_avg}$ is the open-loop observation RMSE averaged across all rollout steps. **Lower CROF = better checkpoint**; all four normalized terms are oriented so that smaller is better.

Rationale. ROF is the primary signal: low ROF means the reward predictor reads from latent directions that need no observation grounding — exactly the regime open-loop CEM-MPC and latent-imagination A2C operate in. The three regularizers protect against degenerate corners of that signal: $(1 - k_c)$ penalizes low controllability rank (insufficient action influence on the latent state), $(1 - k_o)$ penalizes low observability rank (uninformative encoder/decoder pathway), and e_{obs} penalizes high open-loop prediction error (sanity check that the dynamics are reasonable).

Two weight variants. The two variants probe how the relative weight of the dominant `jac_rof_combined` term against the three auxiliary regularizers affects checkpoint selection. CROF-A weighs all four terms equally, whereas CROF-B halves the regularizer weights so the ROF signal carries proportionally more of the score. We report both as a transparent ablation rather than picking a single recipe a priori; the resulting trade-off between selection sharpness and global correlation with MPC return is quantified empirically in Sections 5.2 and 5.6.

Name	Category	What it measures
val_loss, val_kl, val_recon	Loss	Total/KL/reconstruction val. loss
post_obs_rmse, post_rew_rmse	Loss	1-step posterior obs/reward RMSE
ol_obs_start/avg/max/end	Open-loop	Obs RMSE at various horizons
ol_rew_start/avg/max/end	Open-loop	Reward RMSE at various horizons
ol_cumrew_err	Open-loop	Abs. cumulative reward error
jac[_dyn]_spec_radius	Jacobian	Local stability of dynamics
jac[_dyn]_ctrl_rank/cond	Jacobian	Controllable dims / conditioning
jac[_dyn]_obs_rank/cond	Jacobian	Observable dims / conditioning
jac[_dyn]_rcf/ocf/rof	Jacobian	Subspace alignment fractions (good states)
jac_rof_bad, jac_dyn_rof_bad	Jacobian	ROF on failure (low-return) states
jac_rof_combined	Jacobian	$0.5 \cdot \text{jac_rof} + 0.5 \cdot \text{jac_rof_bad}$
jac_co_overlap	Jacobian	Ctrl-obs subspace overlap
emp_C, emp_O, emp_L	Empirical	Action sens., state sens., max-Lipschitz
CROF-A, CROF-B	Composite	Weighted ROF + rank + OL error

Table 1: Summary of all metrics computed per checkpoint. The `jac_dyn_` prefix denotes time-varying linearization; `jac_` denotes fixed linearization.

4 Experimental Setup

4.1 Environment

We use Gymnasium’s LunarLander-v3, a 2D control task where the agent must land a spacecraft between two flags on the lunar surface. The environment provides 8-dimensional observations (position, velocity, angle, angular velocity, leg contacts) and 4 discrete actions (NOP, left thruster, main engine, right thruster). The reward function combines shaping rewards for proximity to the landing pad, speed, orientation, and leg contact, with large bonuses for successful landing (+100 per leg) and penalties for crashing or excessive fuel use.

4.2 Dataset Collection

We collected 872 episodes (158,685 training transitions plus 22,231 validation transitions; 180,916 environment steps in total) through manual human control of LunarLander-v3, split into 750 training episodes (32,038 valid sequence positions at sequence length 30, stride 5) and 122 validation episodes (4,500 positions). Two human pilots took turns flying the lander in order to introduce stylistic diversity, and we deliberately kept a mix of clean landings, near-misses, low-quality flights, and a small number of crashes so that the world model would see both reward-positive and reward-negative trajectory regimes. This dataset is collected once and used for all subsequent world model

training, actor-critic training, and metric evaluation — no additional environment interactions are needed for any model-based method in this paper.

4.3 World Model Training

The RSSM world model (WM) is trained for 500 epochs on the training dataset with: learning rate 10^{-4} (AdamW), batch size 64, sequence length 30 with step offset stride 5, KL free-bits floor $\beta_{\text{KL}}=0.5$, and loss weights as specified in (2). Checkpoints are saved every 5 epochs, yielding 100 checkpoints for evaluation. Unless stated otherwise, all random seeds in this paper (dataloader shuffling, MPC and A2C evaluation episode selection, metric sampling, model initialization) are fixed to seed = 12345, so that any two methods or two checkpoints that report numbers in this paper are evaluated on identical random draws.

4.4 MPC Evaluation

Each of the 100 world model checkpoints (epochs 5, 10, $\dots$, 500) is evaluated using CEM-MPC (Section 3.2) with deterministic seeding for 20 episodes per checkpoint. We report the mean return across these 20 episodes. CEM parameters: horizon $H=25$, population $N=384$, elites $E=48$, iterations $I=4$, smoothing $\alpha=0.7$, planning discount $\gamma=0.97$.

4.5 Metrics Evaluation

All metrics from Section 3.4 are computed for each of the 100 checkpoints on the validation set (122 episodes, 4,500 valid sequence positions at length 30 stride 5; stride is reduced to 1 for the curated good/bad subsets, yielding 2,713 “good” and 1,027 “bad” candidate sequences). Jacobian analysis uses $N_J=128$ sampled latent states per checkpoint (per state-class for the curated metrics), each collected after a 5-step posterior warm-up, with a 25-step rollout horizon matching the CEM planning horizon. A fixed-seed dataloader makes every checkpoint see the same validation states for sampling-based metrics.

Curated good/bad sampling. Each validation sequence is annotated with the total return of its parent episode. We use thresholds $R_{\text{good}} \geq +100$ and $R_{\text{bad}} \leq -100$ to define the two state classes; standard (non-curated) metrics are computed over the full validation set, and ROF is additionally computed twice (once per state class) to produce `jac_rof`, `jac_rof_bad`, `jac_dyn_rof`, and `jac_dyn_rof_bad`.

4.6 Actor-Critic Training

To test whether structural selection criteria translate into downstream actor-critic policies, we train one world-model-based A2C agent per world-model (WM) checkpoint, sampled by a *purposive* (not uniform) protocol that maps directly onto the selection rules of Table 5: every selection rule we report is paired with an actual A2C training outcome. The nine WM checkpoints used are:

- **Metric-driven picks:** WM 310 (smoothed-MPC oracle), WM 280 (raw min CROF-A and CROF-B), WM 315 (smoothed min CROF-A), WM 300 (plateau midpoint, included to test plateau smoothness), and WM 250 (raw min `jac_rof_combined` — pure-ROF baseline without the CROF auxiliary terms).
- **Pre-plateau reference:** WM 200.

- **Early-training references:** WM 50, WM 60, WM 100. WM 60 in particular is the raw single-checkpoint MPC peak (+173.5, but a noisy outlier; see Section 5.1); it is included to test whether raw-MPC spikes survive A2C training.

A uniform sweep over all 100 WM checkpoints would be prohibitive (each WM costs an A2C training run plus a 100-episode evaluation per save); this 9-point purposive design covers early/pre-plateau/plateau and recovers every selection rule of Table 5 at a fraction of the budget.

All nine A2C runs use identical hyperparameters (Section 3.3): training for 1000 epochs with checkpoints every 50 epochs, with all seeds fixed (12345). A2C performance is evaluated by running each saved actor checkpoint in the real LunarLander-v3 environment for 100 deterministic episodes (with seed sequence $12345+k$, $k = 0, \dots, 99$, max 600 steps per episode) — matching the official LunarLander-v3 “solved” criterion of mean return $\geq +200$ over 100 consecutive episodes — and reporting mean return, worst-case return, catastrophic failures (return < -100), and perfect landings (return $\geq +200$).

4.7 Model-Free Baseline

For data-efficiency comparison, we train a model-free Advantage Actor-Critic (A2C) agent that interacts directly with the real environment. The A2C uses observation-space actor and critic networks (3-layer MLPs, hidden dimension 256) and trains for 1000 epochs with 50 episodes per epoch (50,000 total real episodes, ≈ 18.4 M environment transitions, mean episode length ≈ 368 steps), learning rate 3×10^{-4} , $\gamma=0.99$, $\lambda=0.95$, entropy coefficient decaying from 0.2 to 0.01.

5 Experimental Results

5.1 MPC Performance Over Trained World Model Checkpoints

Figure 2 shows CEM-MPC performance across all 100 world model checkpoints. Per-checkpoint mean returns are highly variable (std ≈ 80 –120 across 20 episodes, dominated by the stochastic initial conditions of LunarLander), so we treat the 7-point moving-average curve as the more robust target for selection. We found three training regimes:

- **Early-training / improving:** epochs earlier than ~ 200 . Epoch 60 achieves mean return +173.5, but this is a noisy outlier (worst-case return -170.5); both the surrounding 5-epoch neighborhood and the smoothed curve are far below this. The 7-point moving average (MA-7) centered at epoch 60 is only +81.7, and the immediate neighbors (epochs 55 and 65) score only +55.4 and +29.9 respectively. With MA-7 smoothing, each smoothed point pools $7 \times 20 = 140$ episodes, which is sufficient for using the smoothed (MA-7) MPC curve as the oracle for selection.
- **Smoothed peak:** the 7-point moving average peaks at epoch 310 (smoothed return +153.0), with a broad plateau of high-quality checkpoints spanning epochs ~ 260 – ~ 320 . Worst-case per-episode returns inside the plateau remain noisy at $n=20$, so we treat the smoothed mean as the proper oracle and the plateau as a neighborhood rather than a single point. Selecting the right checkpoint inside this plateau is exactly what CROF is designed to do offline.
- **Late-training collapse:** after epoch ~ 380 the smoothed return drops sharply, with several checkpoints in the epoch 395–465 range exhibiting strongly negative returns (e.g., epoch 405: -55.4). This collapse is invisible to standard training metrics (validation loss, multi-step RMSE), which all keep improving monotonically.

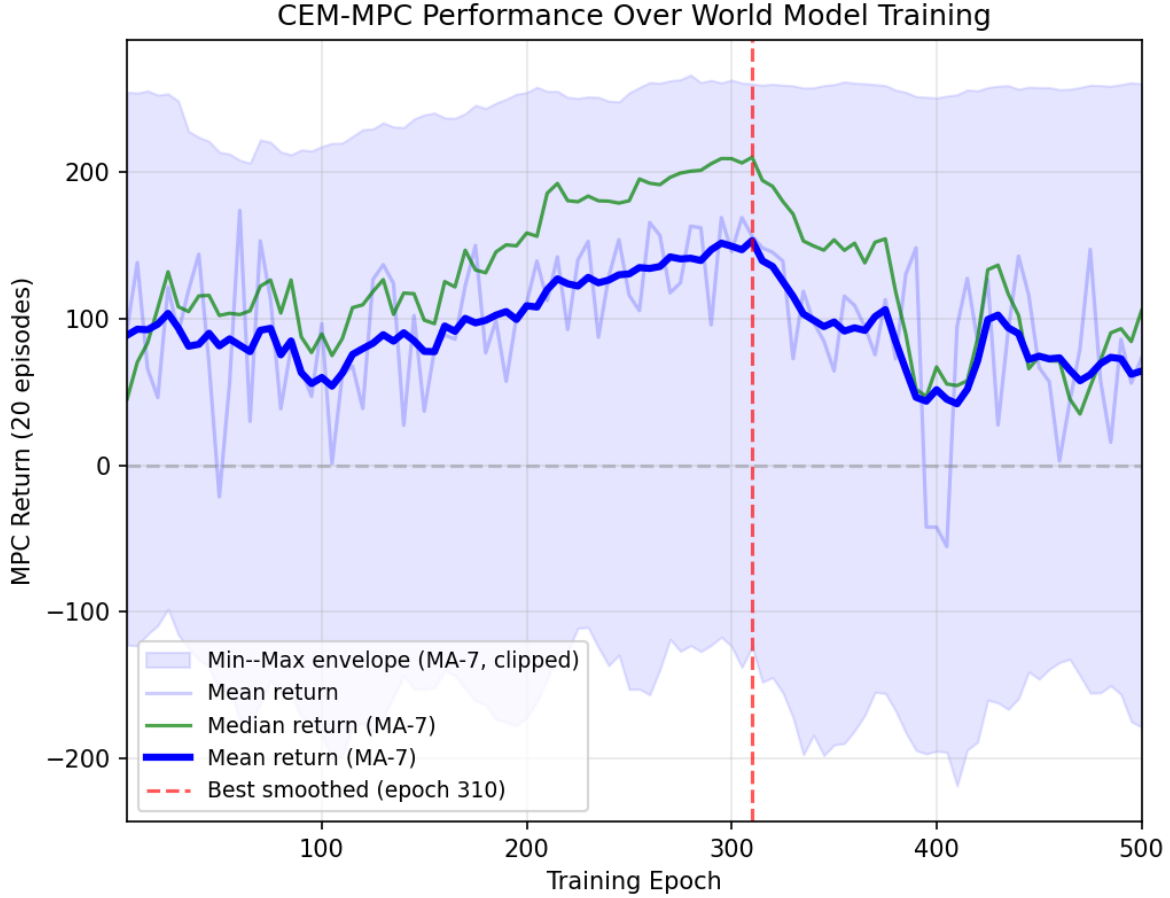


Figure 2: CEM-MPC performance over 500 epochs of trained world model checkpoints using 20 test episodes per checkpoint. With MA-7 smoothing, each smoothed point pools $7 \times 20 = 140$ episodes, which is sufficient for identifying the high-MPC plateau while keeping the 100-checkpoint sweep computationally tractable (a full $n=100$ per-checkpoint sweep would require ~ 12 days). Bold blue: mean return smoothed with a 7-point moving average (MA-7); faint blue: raw mean. Green: median return smoothed with the same MA-7 filter. Light blue band: smoothed min-max envelope; a few extreme outlier points are clipped by the panel limits. The MA-7 mean peaks around epoch 310 and degrades sharply after epoch 380.

5.2 Metric Correlation Analysis

We compute Pearson r , Spearman ρ_s , and quadratic R_q^2 for each metric against the smoothed MPC mean return (7-point moving average). Table 2 shows the strongest predictors and a representative selection of weak ones; the visual ranking restricted to metrics that meet a $|\rho_s| \geq 0.10$ relevance threshold is shown in Figure 3 (the remaining metrics all have $|\rho_s| < 0.10$ and are effectively non-predictive).

Metric	Pearson r	Spearman ρ_s	R_q^2
jac_rof_combined	-0.683	-0.710	0.520
jac_rof (fixed, good states)	-0.558	-0.648	0.479
jac_rof_bad (fixed, bad states)	-0.656	-0.692	0.434
CROF-B	-0.495	-0.568	0.421
CROF-A	-0.375	-0.443	0.243
jac_dyn_rof (time-varying)	-0.467	-0.477	0.229
jac_dyn_rof_bad	-0.277	-0.395	0.182
jac_dyn_spec_radius_max	-0.068	-0.085	0.161
emp_L (max-Lipschitz)	+0.327	+0.154	0.110
ol_rew_end	+0.272	+0.301	0.101
val_loss	+0.029	+0.090	0.015
val_recon	-0.082	-0.184	0.018
post_obs_rmse	-0.084	+0.012	0.010
ol_obs_avg	-0.087	+0.017	0.011
ol_obs_end	-0.097	-0.011	0.013
ol_cumrew_err	+0.002	+0.107	0.062
emp_C	+0.069	-0.093	0.029
emp_O	+0.104	+0.130	0.051

Table 2: Correlation of selected metrics with smoothed MPC return (MA-7) for $N=100$ checkpoints out of 40 metrics computed. The table is split by the horizontal midrule into the **top-10 strongest predictors** (all with both $R_q^2 \geq 0.10$ and $|\rho_s| \geq 0.10$) and a **selection of 8 weak / near-zero predictors** drawn from the remaining metrics whose correlations are all near the noise floor. The full ranking restricted to $|\rho_s| \geq 0.10$ (21 metrics) is shown in Figure 3. **jac_rof_combined** combines **jac_rof** and **jac_rof_bad** and dominates all other metrics on every correlation measure. Both CROF variants and the per-class ROF sub-metrics are strong secondary predictors. Standard training metrics (**val_loss**, multi-step RMSE, action/state sensitivity **emp_C**/**emp_O**) all have near-zero predictive power; the empirical max-Lipschitz **emp_L** is the only sensitivity metric to clear the relevance threshold, but only marginally ($R_q^2=0.110$).

5.3 Fixed vs. Time-Varying Jacobian Analysis

Contrary to our initial expectation, the *fixed* linearization is the stronger MPC predictor across the board. Table 3 shows the head-to-head comparison. For the key ROF metric, **jac_rof** (fixed, good states) achieves $\rho_s = -0.65$ and $R_q^2=0.48$ versus $\rho_s = -0.48$ and $R_q^2=0.23$ for **jac_dyn_rof**, and the gap on the bad-state subscore is even larger ($\rho_s = -0.69$ vs. -0.40).

Reasons for the fixed-Jacobian win. One would expect time-varying linearization to better capture the non-linear dynamics actually experienced during a 25-step CEM-MPC rollout. We see three plausible reasons for the empirical reversal in our LunarLander setting: (i) *long-horizon*

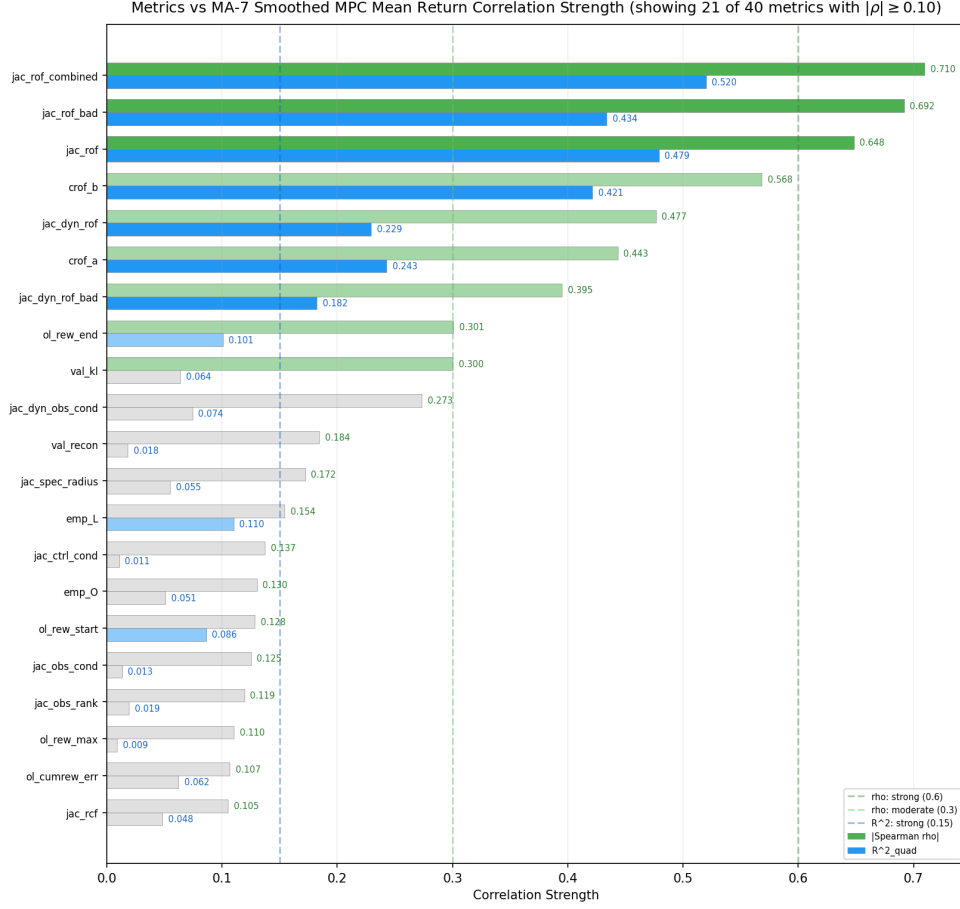


Figure 3: The 21 (of 40 total) metrics with $|\rho_s| \geq 0.10$ ranked by Spearman $|\rho_s|$ (light green) and quadratic R_q^2 (blue) against smoothed MPC return. The 19 metrics omitted from the chart all have $|\rho_s| < 0.10$ and are effectively non-predictive; representative examples from this noise floor are listed below the midrule of Table 2. **jac_rof_combined**, the per-class ROFs, and CROF metrics dominate; all standard training metrics cluster at the lower end. The empirical max-Lipschitz **emp_L** is the only finite-difference sensitivity metric to clear the threshold, sitting in the middle of the chart.

Sub-metric	Fixed (jac_-)			Time-varying (jac_dyn_-)			Winner
	r	ρ_s	R_q^2	r	ρ_s	R_q^2	
spec.radius	-0.157	-0.172	0.055	-0.080	-0.087	0.032	fixed
ctrl.rank	+0.113	+0.015	0.013	+0.059	-0.071	0.038	dyn
obs.rank	+0.107	+0.119	0.019	+0.048	-0.028	0.030	dyn
rcf	-0.194	-0.105	0.048	-0.154	-0.023	0.029	fixed
ocf	-0.121	+0.054	0.015	-0.147	-0.031	0.049	dyn
rof	-0.558	-0.648	0.479	-0.467	-0.477	0.229	fixed
rof_bad	-0.656	-0.692	0.434	-0.277	-0.395	0.182	fixed

Table 3: Head-to-head: fixed vs. time-varying Jacobian sub-metrics against smoothed MPC return. On the most informative sub-metrics (the two ROF variants in bold), the fixed linearization wins by a wide margin in both Spearman ρ_s and quadratic R_q^2 .

error compounding — the time-varying $\mathcal{O}_H$ multiplies $H-1$ Jacobians along an uncorrected open-loop rollout, so small per-step linearization errors compound multiplicatively while the fixed A^k formulation is biased but lower-variance; (ii) *trajectory mismatch* — CEM re-plans every step rather than traversing the open-loop trajectory used to compute A_k, B_k, C_k , so the fixed Jacobian at the current state is closer to “what CEM sees”; and (iii) *sample regime* — at $N_J=128$ and $H=25$ the time-varying formulation places its 3,200 Jacobians at possibly drifting states, while the fixed variant uses only 128 Jacobians at warm-up-grounded states, a cleaner signal at our sample budget. We therefore do not use the time-varying metrics for our composite CROF scores, since they double evaluation cost without improving the predictive signal; they are still available in the released code (enabled with `--compute_dyn_jac`, default off).

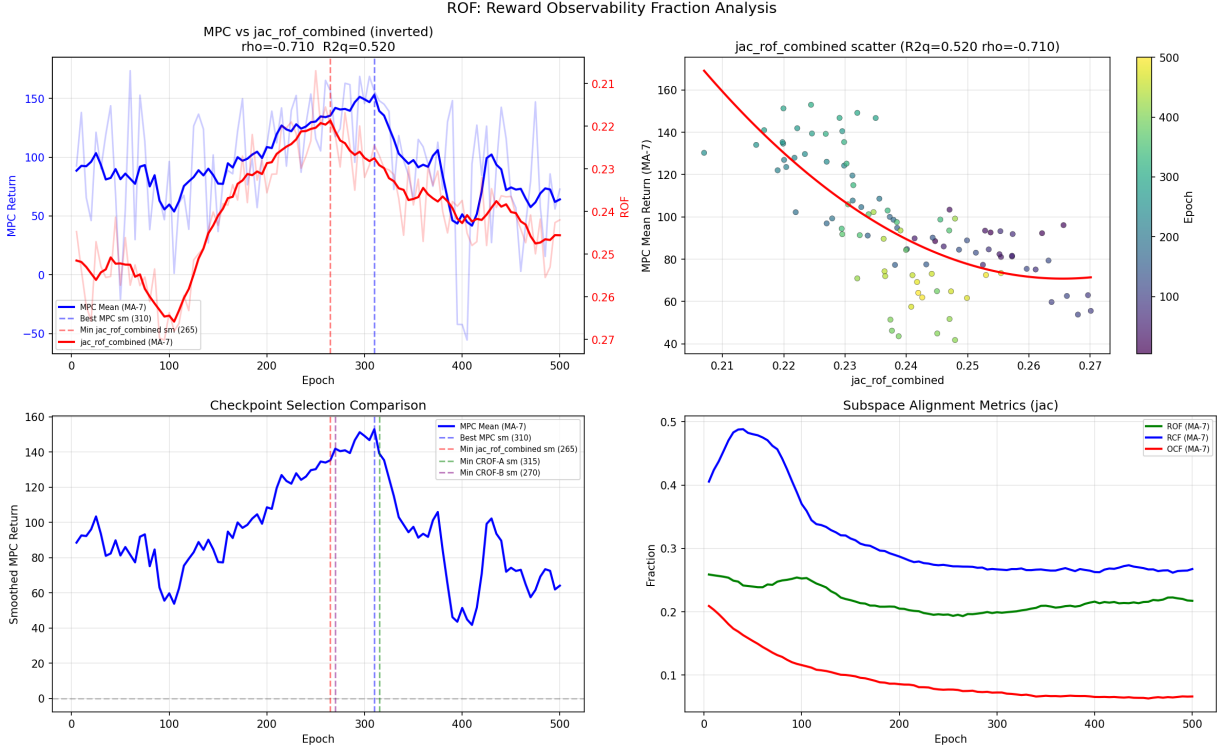


Figure 4: ROF analysis. Top-left: smoothed MPC return (blue) vs. inverted `jac_rof_combined` (red), with vertical lines at the smoothed MPC peak (epoch 310) and the smoothed `jac_rof_combined` minimum (epoch 265). Top-right: per-checkpoint scatter with quadratic fit, exhibiting the characteristic U-shape (smoothed Spearman $\rho_s = -0.71$, $R_q^2 = 0.52$). Bottom-left: checkpoint selection comparison — the MA-7-smoothed `jac_rof_combined` minimum and the MA-7-smoothed CROF-A and CROF-B picks all sit inside or adjacent to the high-MPC plateau (raw minima in Table 5). Bottom-right: subspace alignment metrics (RCF, OCF, ROF) vs. epoch.

5.4 ROF as a Predictor of MPC Performance

`jac_rof_combined` is the strongest single co-varying metric in our suite on this LunarLander setup. Its Spearman rank correlation $\rho_s = -0.71$ against smoothed MPC return is statistically robust ($N=100$ checkpoints, $p \lesssim 10^{-15}$), and its quadratic $R_q^2 = 0.52$ is the largest in the table. The two per-class subscores are individually strong as well (ρ_s of -0.65 for good states and -0.69 for bad

states), so the combined score rests on two largely independent pieces of evidence rather than a single regression artefact.

Mechanism. ROF measures how much of the reward gradient’s energy lies in the observable subspace. During training, the posterior update anchors observable directions using real observations. MPC planning is open-loop: the model is rolled forward for H steps from the prior alone, so over a 25-step horizon the observable directions drift uncorrected. Low ROF means the reward predictor reads from latent directions that need no observation grounding and therefore remain accurate through the rollout. High ROF means it reads from the same drifting features the decoder uses, making reward predictions unreliable. ROF is therefore a measure of how much of the reward signal sits in the part of the latent that requires observation grounding yet receives none during open-loop imagination. Section 5.5 provides a separate, model-free check that this matters for LunarLander specifically: $\sim 71\%$ of per-step reward variance is unrecoverable from (o_t, a_t) alone, so a well-trained world model must move that part of the reward computation out of the observation-decoder column space. This is the regime ROF is designed to detect.

Curated-sampling effect. The strong correlation in Table 2 relies on the fixed 50/50 good/bad composition described in Sections 3.4 and 4.2; ROF on uniformly sampled states correlates much more weakly with MPC return.

5.5 Reward Predictability from Observations

To check whether the ROF interpretation is mechanistically grounded for LunarLander, we ran a small static-regression experiment outside the world-model pipeline. The question is: how much of the per-step reward r_t is recoverable from observations alone? We trained MLP regressors (two hidden layers of 256 units, ReLU, AdamW, MSE loss, early stopping on validation R^2) on the same offline dataset used for world-model training, with four input configurations: (o_t, a_t) and (o_t, a_t, o_{t+1}) , each evaluated with terminals kept and with terminals filtered (done=1 rows account for $<1\%$ of training data). Results in Table 4.

Config	Input	Terminals	val R^2
A	(o_t, a_t)	kept	0.290
B	(o_t, a_t)	filtered	0.670
D	(o_t, a_t, o_{t+1})	kept	0.595
C	(o_t, a_t, o_{t+1})	filtered	0.971

Table 4: LunarLander per-step reward predicted by a static MLP. The single-step input the world-model reward head sees from a one-step perspective (config A) leaves $\sim 71\%$ of variance unexplained. Two structural causes account for the gap: a per-step shaping term that references o_{t+1} (closes $+0.30$ when added) and terminal ± 100 events that depend on Box2D-internal flags absent from any observation (closes $+0.38$ when filtered). Together they explain the gap up to a small dispersion-noise floor (see text).

The headline number is that the observation–action pair (o_t, a_t) leaves $\sim 71\%$ of LunarLander’s per-step reward variance unexplainable. Both causes are visible in the gym source. (i) The shaping component is a potential difference of the form $\phi(o_{t+1}) - \phi(o_t)$, which strictly cannot be computed from o_t alone — adding o_{t+1} recovers ~ 30 percentage points of R^2 . (ii) Terminal ± 100 events are gated by simulator-internal flags (`lander.awake`, `game_over`, leg ground-contact bits) that are

not exposed in the 8-D observation — filtering terminals recovers another ~ 38 percentage points. The remaining $\sim 3\%$ residual in the best (full-input, terminals-filtered) regressor is consistent with the engine-impulse dispersion noise that Box2D injects at every action step (a small random offset applied to the impulse position each frame).

This connects directly back to ROF. A sequence-model latent state can in principle track the missing simulator flags from trajectory history, and a well-trained reward head therefore must read from latent directions that lie outside the observation-decoder column space. ROF is, by construction, the fraction of the reward gradient’s energy that does sit inside that column space, so as the world model internalizes the missing simulator state, ROF should decrease. Sections 5.2 and 5.4 document this empirically: `jac_rof_combined` drops along the high-MPC region of training and rises back as the model overfits. We do not claim this mechanism transfers to environments whose reward is fully Markovian in (o_t, a_t) ; on those tasks we would expect ROF to increase as training progresses.

5.6 Checkpoint Selection via CROF

Table 5 evaluates checkpoint selection using different criteria. We report two oracle baselines (raw and smoothed best MPC return), both raw and smoothed-min selections for the structural metrics, and representative weak baselines.

Selection rule	Epoch selected	MPC return
<i>Oracle baseline</i>		
Best smoothed MPC (oracle, MA-7)	310	+153.0
<i>Structural / proposed metrics</i>		
Min CROF-A (raw)	280	+162.8
Min CROF-A (smoothed MA-7)	315	+148.1
Min CROF-B (raw)	280	+162.8
Min CROF-B (smoothed MA-7)	270	+117.5
Min <code>jac_rof_combined</code> (raw)	250	+115.8
Min <code>jac_rof_combined</code> (smoothed MA-7)	265	+156.5
<i>Standard / weak baselines</i>		
Min <code>val_loss</code>	470	+79.3
Min <code>post_rew_rmse</code>	470	+79.3
Min <code>ol_obs_end</code>	495	+55.9
Min <code>ol_cumrew_err</code>	460	+2.9
Max <code>emp_C</code>	485	+15.6
Max <code>jac_ctrl_rank</code>	445	+115.8

Table 5: Checkpoint selection comparison. The smoothed (MA-7) MPC oracle peaks at epoch 310 with a plateau over epochs 260–320; both CROF variants’ raw-minimum picks (epoch 280) land inside it. The pure-ROF baseline (`jac_rof_combined`) picks epoch 250 raw / 265 smoothed — only the smoothed pick is in the plateau, motivating the CROF auxiliary terms. All standard criteria pick deeply overfit checkpoints (epoch ≥ 460) with MPC returns of +3 to +79.

A few observations are worth making explicit:

- The raw ROF signal (`jac_rof_combined`) has higher pure correlation with MPC return but its raw minimum sits at epoch 250, earlier in the high-MPC plateau. This is the motivation for the composite score: the regularizer terms in CROF nudge the selected epoch into the high-MPC plateau.

- Standard selection criteria are poor. Validation loss, posterior reward RMSE, multi-step observation RMSE, and cumulative-reward RMSE all monotonically improve through training (Figure 6), so all pick badly overfit checkpoints (epoch ≥ 460) where MPC has collapsed.
- Among the metrics we tested, ROF and CROF are the ones that track MPC’s rise–peak–decline pattern: a phase-wise breakdown shows `jac_rof_combined` reaches Pearson $r = -0.65$ in mid-training (epochs 100–300) versus -0.20 early and -0.32 late — ROF is most informative precisely in the regime where one would actually select a checkpoint.

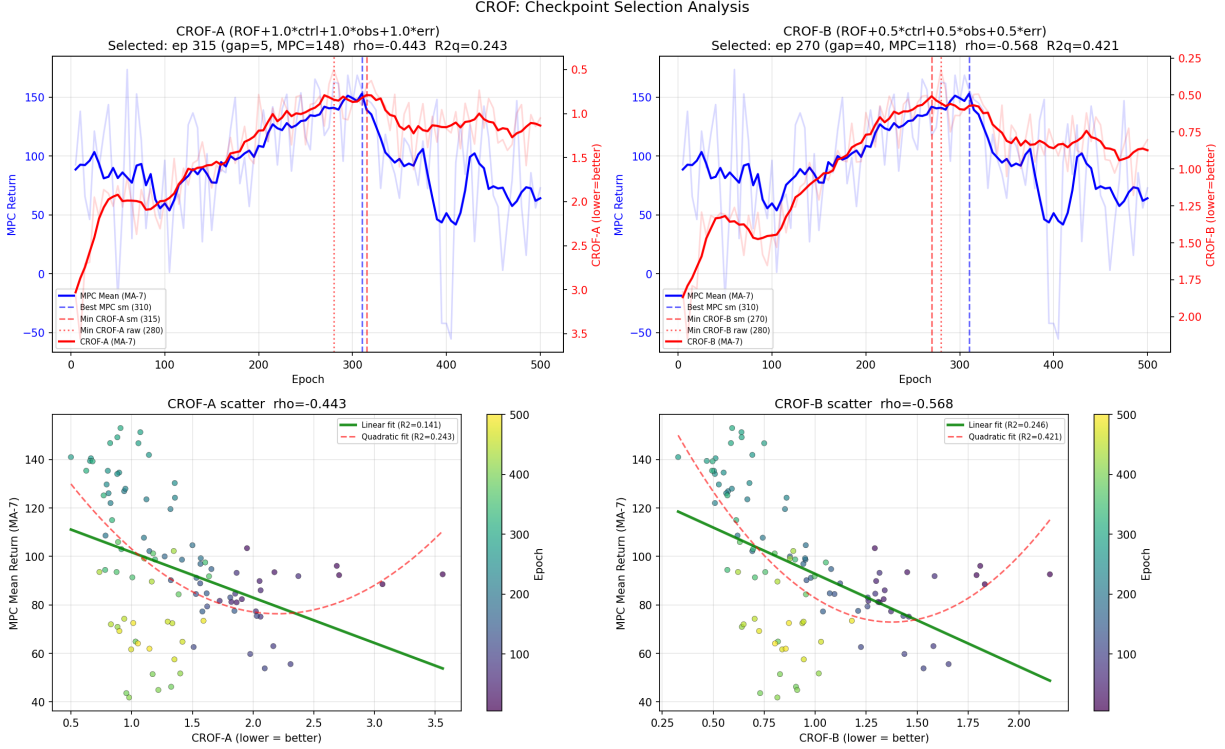


Figure 5: CROF composite score analysis. Top row: smoothed MPC return (blue) vs. smoothed CROF-A and CROF-B (red, lower-is-better axis inverted). Vertical lines mark the smoothed MPC oracle (blue dashed, epoch 310), the MA-7-smoothed CROF minimum (red dashed: CROF-A at 315, CROF-B at 270), and the raw CROF minimum (red dotted: 280 for both variants); see Table 5. Bottom row: per-checkpoint scatter with linear (green solid) and quadratic (red dashed) OLS fits coloured by epoch. A few high-CROF, low-return early-epoch outliers pull the OLS fits and create a spurious “valley”; the Spearman ρ in each panel title is the robust single-number summary and confirms the monotonic “lower CROF = higher MPC return” relationship.

5.7 Structural Insights

The Jacobian analysis reveals several structural properties:

Low-dimensional controllability. The controllability rank plateaus at $k_c \approx 33\text{--}47$ (depending on fixed vs. time-varying) out of $n = 272$ total latent dimensions. Only $\sim 12\text{--}17\%$ of the latent space is reachable by action sequences over the 25-step planning horizon. The GRU’s gating mechanism preserves historical context in the remaining dimensions, making them effectively uncontrollable.

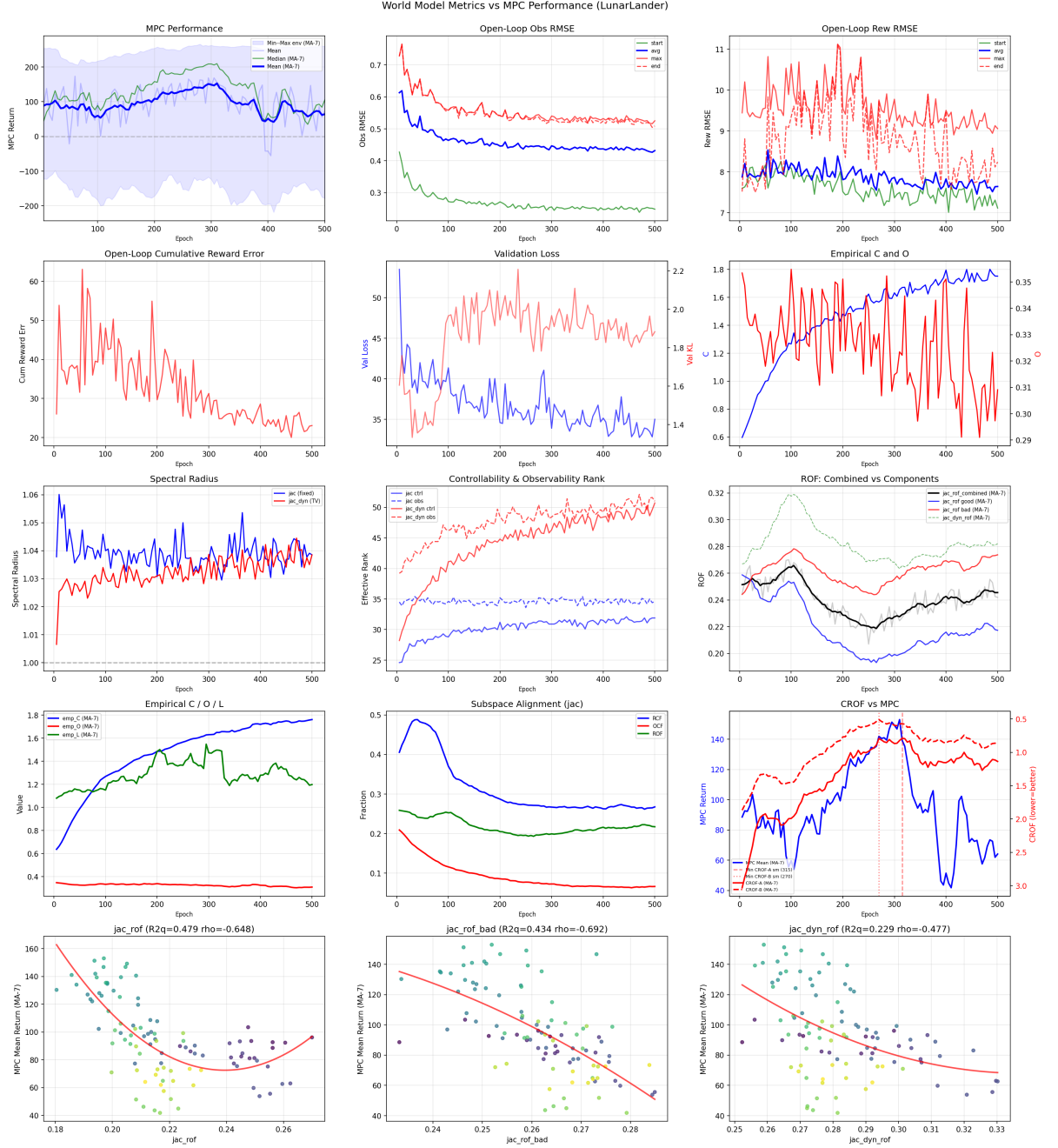


Figure 6: Comprehensive metrics vs. MPC performance dashboard. Top four rows: time series of standard metrics (validation losses, open-loop RMSEs, empirical sensitivities, Jacobian spectral radius and ranks); almost all are monotonic or flat through the late-training collapse and provide no signal for checkpoint selection. The third-row right panel shows the ROF family (**jac_rof_combined** bold black, **jac_rof** good blue, **jac_rof_bad** red, **jac_dyn_rof** dashed green) exhibiting a rise–peak–collapse U-shape that mirrors MPC’s trajectory. Bottom row: scatter of the three core ROF sub-metrics against MA-7 MPC return with quadratic fits.

Overfitting is structural, not statistical. The central finding is that world model overfitting manifests in the *geometry* of the latent space rather than in prediction accuracy. Multi-step prediction RMSE continues to improve. The model keeps getting better at predicting future states, but the structural relationship between rewards, observations, and controllable dimensions becomes distorted which hurts closed-loop MPC performance.

5.8 Impact on Model-based Actor-Critic Policy Training Performance

Having seen that CROF identifies world-model checkpoints inside the high-MPC plateau on this LunarLander setup, we now ask whether this selection also translates into better downstream actor-critic training. For each of the nine world-model (WM) checkpoints described in Section 4.6 we trained a world-model-based actor-critic (A2C) policy for 1000 epochs (saving a checkpoint every 50 epochs, with identical hyperparameters across all runs) and evaluated every saved actor checkpoint in the real environment for 100 deterministic episodes (seed sequence $12345+k$, $k = 0, \dots, 99$; max 600 steps). Table 6 reports, per used WM checkpoint: (i) the A2C checkpoint with the highest mean return (“best-by-mean”), and (ii) the A2C checkpoint with the highest worst-case return (“safest-by-worst”) — the more practically useful selection when no real-environment evaluation budget is available.

WM	Selection rule	Best-by-mean A2C save					Safest-by-worst A2C save					#safe
		Ep.	Mean	Worst	Catas	Perfect	Ep.	Mean	Worst	Catas	Perfect	
50	early training	550	+178.3	−120.8	5	64	900	+141.3	−74.7	0	47	1/20
60	raw-MPC outlier	500	+206.3	−157.3	5	79	600	+182.1	−154.7	8	73	0/20
100	early training	800	+223.3	−108.0	1	85	200	+65.0	−80.2	0	25	3/20
200	pre-plateau reference	1000	+191.4	−184.2	3	70	300	+149.1	−111.3	1	43	0/20
250	min jac.rof.combined (raw)	300	+141.0	−161.2	1	26	200	+127.3	−104.8	1	28	0/20
280	min CROF-A, CROF-B (raw)	800	+217.5	−178.9	2	79	300	+191.3	−39.6	0	71	2/20
300	plateau midpoint	850	+205.9	−95.5	0	74	600	+138.3	−63.1	0	49	7/20
310	smoothed-MPC oracle	1000	+209.6	−131.0	2	75	400	+196.0	−40.4	0	52	3/20
315	min CROF-A (sm. MA-7)	800	+178.0	−203.5	1	63	700	+133.1	−72.1	0	47	1/20

Table 6: A2C training outcomes across purposively sampled world-model checkpoints, 100 deterministic episodes per A2C save (bold rows mark CROF metric-driven picks). **Ep.** is the AC training epoch of the reported save; **mean/worst** are mean and worst-case return; **catas** counts episodes with return < -100 , **perfect** counts $\geq +200$; **#safe** is the number of A2C saves (out of 20) with 0 catastrophic episodes. The *best-by-mean* (BBM) and *safest-by-worst* (SBW) blocks report, respectively, the A2C save with the highest mean and the save with the highest worst-case return. A deployable WM is one whose BBM mean and SBW mean are both high (small peak-to-safe gap). By that criterion the CROF raw pick **WM 280** dominates: BBM +217.5 and SBW +191.3 with a ~ 26 -point peak-to-safe gap, against ~ 158 for the highest-BBM trap WM 100. The smoothed-MPC oracle (WM 310) is comparable to WM 280 but requires the full MPC sweep to identify.

Offline-only selection. On LunarLander, CROF lands inside the cluster of the best and safest world-model checkpoints for model-based A2C training, using validation-time diagnostics only, with no real-environment evaluation in the selection loop. Both CROF picks (WM 280 raw, WM 315 smoothed) sit in the high-quality cluster of Table 6. The smoothed-MPC oracle (WM 310) is comparable but requires the full MPC sweep to identify, which CROF avoids. We have not tested whether the same selection rule transfers to other environments.

CROF vs. pure ROF. WM 250 is the raw minimum of `jac_rof_combined`, i.e. ROF alone, without CROF’s auxiliary controllability, observability, and open-loop-observation-error terms. It is the weakest deployable WM in the sweep (BBM +141.0, SBW +127.3). The auxiliary terms are not cosmetic: they nudge the selection out of the local-ROF dip at WM 250 and into the structurally good region around WM 280 where A2C training is reliable.

Best-by-mean is misleading. The four early/pre-plateau references (WMs 50, 60, 100, 200) all reach high BBM means but fail the worst-case test (≥ 1 catastrophic at their BBM save). WM 100 is the trap: highest BBM (+223.3, 85/100 perfect), but its safest save drops to mean +65.0. That is a ~ 158 -point peak-to-safe gap, against ~ 26 for WM 280. The safest-by-worst column is where structurally good WMs separate from the noisy ones.

5.8.1 Model-free A2C

We also trained a model-free A2C (MF-AC) agent to compare against the world-model-based A2C (WM-AC) policies (Section 4.7). MF-AC is trained for 1000 epochs, saving every 50th checkpoint. Each saved checkpoint is evaluated with the same 100-episode protocol used for WM-AC (seed sequence $12345+k$, $k = 0, \dots, 99$). To give MF-AC the strongest possible evaluation budget, we use the standard `LunarLander-v3` horizon of 1000 steps per episode; WM-AC and MPC are capped at 600 steps to match MPC’s expensive per-step inference budget. MF-AC peaks at AC epoch 760 with mean return **+193.04** (54/100 perfect landings, **0/100** catastrophic, worst -35.55), and forms a reliable plateau over epochs ~ 610 – 910 in the $+170$ to $+195$ band with 0/100 catastrophic episodes at all but one checkpoint. The head-to-head against the WM-AC picks is reported in Table 7: MF-AC has the tightest worst-case at its peak save, while the WM-AC picks have higher mean and more perfect landings at a fraction of the data cost.

5.9 Data Efficiency Comparison

A central promise of model-based RL is data efficiency. We measure cost in real environment steps (one step = one (s, a, r, s') tuple) so that CEM-MPC, the world-model-based A2C (WM-AC), and the model-free A2C (MF-AC) are all compared on the same axis. The model-free training log records `transitions` per epoch; we sum these up to MF-AC’s best epoch 760 under the 100-episode, 1000-step evaluation protocol. All model-based methods share the same offline dataset budget (158,685 train + 22,231 val = 180,916 environment steps; $750 + 122 = 872$ episodes). Table 7 summarizes the result.

Headline. The CROF raw WM-AC pick (WM 280, identified offline with no environment evaluation) reaches peak mean **+217.48** at $\sim 65.3\times$ less real-environment interaction than MF-AC. In other words, a structurally selected world model converts ~ 181 K offline transitions into an actor-critic policy whose peak real-environment return exceeds a model-free A2C trained on 11.8M real steps. The smoothed-MPC oracle (WM 310) gives a comparable WM-AC outcome (+209.6) but only after running the full MPC sweep, which CROF avoids; CEM-MPC on the same WM 310 already reaches **+166.6** at $n=100$ with no policy training. MF-AC’s 1000-epoch run consumes ~ 18.4 M steps end-to-end; the 11.8M figure is the budget through its peak save, which is the relevant comparison when no periodic real-environment evaluation is available. At inference time, the trained A2C policy is $\sim 1000\times$ cheaper to execute than CEM-MPC, which performs 4 iterations of 384 rollouts $\times 25$ steps at every real timestep.

Method	Env steps	Mean	Worst	Perfect	Catast.	Ratio vs. MF-A2C
MPC (smoothed best, WM 310)	180,916	+166.60	-138.52	64/100	3/100	65.3 × fewer
WM-A2C (CROF raw, WM 280, A2C ep 800)	180,916	+217.48	-178.86	79/100	2/100	65.3 × fewer
WM-A2C (CROF-A smooth, WM 315, A2C ep 800)	180,916	+178.01	-203.47	63/100	1/100	65.3 × fewer
Model-free A2C (peak @ ep 760)	11,820,726	+193.04	-35.55	54/100	0/100	1× (baseline)

Table 7: Real environment-step cost vs. deterministic-policy mean return at 100 episodes per checkpoint. MPC and WM-A2C use a 600-step cap matching MPC’s inference budget; MF-A2C uses the standard 1000-step **LunarLander-v3** horizon under which it was trained. All model-based methods reuse the same 872-episode offline dataset (180,916 env. steps). **Perfect** counts return $\geq +200$ (LunarLander’s “solved” threshold); **Catast.** counts return < -100 . For each WM-A2C row we report the BBM A2C save inside the corresponding CROF-selected world model (Table 6). The CROF raw pick (**WM 280**) is the strongest model-based result (+217.48, 79/100 perfect), beating MF-A2C’s peak by ~ 24 points at $\sim 65\times$ fewer real environment steps. MF-A2C is the only method with 0/100 catastrophic and the tightest worst-case, but its data cost is two orders of magnitude higher.

6 Conclusion

We presented a study of validation-time metrics for predicting the closed-loop performance of a learned latent world model, with LunarLander-v3 as the testbed and the *Composite Reward-Observability Fraction* (CROF) as a candidate single-number score for offline checkpoint selection. On this task, CROF tracks both CEM-MPC and model-based A2C performance: its dominant `jac_rof_combined` term reaches Spearman $\rho_s = -0.71$ ($R_q^2 = 0.52$) against smoothed MPC return. All standard model-selection criteria (validation loss, multi-step RMSE, empirical sensitivities) instead pick deeply overfit checkpoints where MPC has already collapsed. We also provided a reward regression experiment to explain why reward observability tracks world model’s structural properties for LunarLander.

On LunarLander-v3 with $\sim 181\text{K}$ offline transitions, the CROF raw pick (WM 280) yields a model-based A2C policy with peak mean return **+217.5** (79/100 perfect landings) — ~ 24 points above the model-free A2C baseline (+193.0 over $\sim 11.8\text{M}$ environment steps), i.e. at $\sim 65\times$ less real-environment interaction. The smoothed-MPC oracle (WM 310, identifiable only via the full MPC sweep) yields a comparable WM-AC peak of +209.6, so on this task CROF matches an oracle pick at zero real-environment-selection cost.

Scope and limitations. All claims here are on a single environment (LunarLander-v3). The mechanistic argument for ROF leans on the specific structure of LunarLander’s reward (per-step shaping plus terminal events gated by simulator-internal flags), so we expect ROF to carry less signal on tasks whose reward is fully Markovian in (o_t, a_t) . We therefore present CROF here as a useful diagnostic for selecting RSSM checkpoints in this study, not as a domain-agnostic model-selection rule.

Future work. Natural extensions include: (i) cross-environment validation on other environments with different rewards, with continuous actions, and on high-dimensional environments; (ii) using ROF-like term as an auxiliary training regularization term to directly optimize world model structure for closed-loop planning; (iii) developing an analogous offline metric for selecting the best A2C policy checkpoint during training, so the final actor save can also be picked without environment interaction; this is an obvious gap in the current pipeline.

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